Design and Development of IoT-Based Smart Maldive Fish Processing System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The Maldive fish processing industry in Sri Lanka and the Maldives largely depends on traditional manual inspection methods to evaluate the quality of raw fish before processing. These methods rely heavily on human judgment to assess freshness, size uniformity, and surface defects, which often leads to inconsistent grading, reduced production efficiency, and increased operational costs. Manual inspection also introduces subjectivity and human error, allowing substandard fish to enter the processing pipeline, which can negatively affect drying efficiency and final product quality.

Although previous research has explored computer vision techniques and IoT-based monitoring systems for seafood processing, most existing solutions focus on isolated aspects such as environmental monitoring or freshness detection. Few studies address automated quality assessment at the raw fish intake stage, where early identification of low-quality fish can significantly improve processing efficiency and product consistency.

To address this gap, this research proposes an **AI-Driven Raw Fish Quality Assessment and Standardization system** that utilizes computer vision and machine learning techniques to automatically evaluate fish quality during the intake stage. High-resolution RGB cameras capture images of fish moving on a conveyor system, while weight sensors measure physical characteristics. Deep learning models such as convolutional neural networks will analyse visual features to detect defects, estimate freshness indicators, and classify fish into standardised quality categories such as Grade A, Grade B, or Reject.

The proposed system aims to improve grading accuracy, reduce reliance on manual inspection, and ensure consistent raw material quality before processing. Expected outcomes include increased processing efficiency, reduced waste, and improved product reliability. From a social and economic perspective, the system can support small- and medium-scale seafood processors by introducing intelligent automation, improving competitiveness in international markets, and contributing to sustainable food production.

Keywords: Computer Vision, Fish Quality Assessment, Deep Learning, Smart Food Processing, Industrial Automation.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
IoT	Internet of Things
RGB	Red Green Blue (Color Image Model)
DFD	Data Flow Diagram
UI	User Interface
API	Application Programming Interface
CPU	Central Processing Unit
GPU	Graphics Processing Unit
YOLO	You Only Look Once (Object Detection Algorithm)
OpenCV	Open Source Computer Vision Library
SDG	Sustainable Development Goals
VOC	Volatile Organic Compounds
SME	Small and Medium Enterprises

1. INTRODUCTION

1.1 Background and Context

The seafood processing industry plays a significant role in global food supply and economic development. Among various processed seafood products, Maldive fish traditionally produced from tuna species represents an important export commodity in Sri Lanka and the Maldives. However, many small and medium-scale processing facilities still rely on traditional processing methods and manual inspection procedures to evaluate raw fish quality before processing. These methods depend heavily on human judgment to determine freshness, detect surface defects, and evaluate the overall physical condition of fish. Manual inspection often leads to inconsistent grading and operational inefficiencies due to human error, fatigue, and subjective evaluation criteria.

In traditional Maldive fish production systems, substandard raw fish may enter the processing pipeline due to inaccurate quality assessment. This can result in uneven drying, increased spoilage rates, and unnecessary energy consumption during drying and smoking operations. Studies on food processing automation highlight that manual quality inspection is one of the major bottlenecks in maintaining consistent product quality in seafood industries [1]. Therefore, improving the accuracy and consistency of raw fish evaluation is essential to ensure better production efficiency and product reliability.

The lack of standardized raw fish assessment impacts multiple stakeholders within the seafood processing supply chain. Small and medium-scale fish processors face increased operational costs and reduced productivity due to inefficient manual inspection practices. Exporters and distributors are also affected because inconsistent product quality can reduce competitiveness in international seafood markets and may fail to meet strict food safety standards. Quality assurance officers rely on manual inspection techniques that are time-consuming and prone to inaccuracies. Ultimately, consumers may receive products with inconsistent quality or reduced shelf life.

According to research on seafood quality evaluation, implementing automated inspection technologies can significantly improve accuracy, reduce labor costs, and enhance overall food safety standards [2]. Consequently, introducing intelligent automation into the raw fish evaluation stage is crucial for improving both industrial productivity and consumer trust.

The global food processing industry is increasingly adopting advanced technologies such as computer vision and machine learning to automate quality inspection processes. Recent studies show that deep learning-based image analysis can achieve high accuracy in detecting food defects and predicting freshness levels compared to manual inspection methods [3]. Despite these technological advancements, traditional fish processing industries in developing regions such as South Asia continue to rely heavily on manual grading systems. Studies examining the dried fish industry in South Asia highlight the lack of technological integration in quality

inspection processes and emphasize the need for automated solutions tailored to small-scale processing environments [4].

Several research studies have proposed computer vision techniques for assessing fish freshness and detecting surface defects. For example, convolutional neural networks have been successfully applied to identify visual indicators of fish spoilage and quality deterioration [5]. Similarly, deep learning models such as YOLO and EfficientNet have shown strong performance in detecting surface defects in seafood products [6].

To address these limitations, this research proposes an **AI-Driven Raw Fish Quality Assessment and Standardization system** that utilizes computer vision techniques to evaluate raw fish quality before processing.

The proposed system will use **high-resolution RGB cameras installed above the intake conveyor** to capture images of fish entering the processing line. Computer vision algorithms will analyze these images to detect visual indicators of freshness, identify surface defects such as bruising or discoloration, and measure physical characteristics such as size and shape.

Deep learning models, including **convolutional neural networks and object detection architectures**, will process the captured image data to automatically classify fish into standardized quality categories such as **Grade A, Grade B, or Reject**. The classification results will then be integrated with an **automated conveyor sorting mechanism**, which directs fish into appropriate processing streams based on their quality grade.

A **real-time monitoring dashboard** will display classification results, quality statistics, and system alerts when substandard fish are detected. This system will help operators maintain consistent grading standards while reducing human error and improving operational efficiency in Maldive fish processing facilities.

The proposed research aligns with several **United Nations Sustainable Development Goals (SDGs)**:

- **SDG 2 – Zero Hunger:** Improving food safety and reducing spoilage in seafood processing.
- **SDG 8 – Decent Work and Economic Growth:** Enhancing productivity and technological innovation within small-scale fish processing industries.
- **SDG 9 – Industry, Innovation, and Infrastructure:** Introducing intelligent automation and AI-driven decision systems into traditional manufacturing processes.
- **SDG 12 – Responsible Consumption and Production:** Minimizing resource waste and improving quality consistency in food production.

This research belongs to the **Autonomous Intelligent Machines and Systems (AIMS)** research cluster and falls within the **Information Technology specialization**. The project combines **artificial intelligence, computer vision, and automated industrial inspection systems** to create a smart quality assessment framework for seafood processing industries.

1.2 Literature Review

Recent advancements in artificial intelligence (AI), computer vision, and machine learning have significantly improved automated quality inspection systems in the food processing industry. Researchers have explored several approaches to assess fish freshness, detect surface defects, and automate grading processes. This section reviews key research contributions related to seafood quality inspection and identifies the limitations of existing systems.

Cheng et al. investigated the use of computer vision techniques to evaluate fish freshness using image-based analysis[3]. Their study applied convolutional neural networks (CNNs) to analyze visual indicators such as texture, color, and surface patterns of fish. The results demonstrated that deep learning models can achieve high accuracy in freshness classification compared to manual inspection methods. However, the study mainly focused on laboratory-based image datasets and did not consider real-time industrial deployment.

Similarly, Liu et al. proposed a computer vision-based system that uses deep convolutional neural networks to assess fish product quality[5]. Their approach extracts visual features from images and classifies fish samples into quality categories. While this method improves classification accuracy, the system relies entirely on image data and does not consider practical deployment challenges such as varying lighting conditions or real-time processing requirements in industrial environments.

Another study conducted by Das et al. explored the use of deep learning models for seafood defect detection[6]. Their system used advanced object detection techniques to identify surface defects such as bruising, discoloration, and physical damage. Although the approach demonstrated promising detection performance, the system was mainly designed for detecting defects rather than evaluating overall fish quality or freshness.

Research by Mir et al. introduced an AI-based system for quality assessment in fish processing industries[2]. The study emphasized the potential of machine learning algorithms in automating inspection tasks and improving processing efficiency. However, the proposed approach focused primarily on predictive quality evaluation and did not integrate automated sorting or real-time processing mechanisms.

In addition, Kulatunga et al. examined the challenges faced by traditional dried fish processing industries in South Asia[4]. The study highlighted the heavy reliance on manual labor for quality grading and emphasized the lack of technological integration in small-scale seafood processing operations. The authors recommended adopting automation and digital monitoring systems to improve productivity and quality consistency.

Wang et al. reviewed machine vision applications in food quality inspection and reported that computer vision systems can significantly improve accuracy and consistency compared to manual inspection methods[7]. However, many of these systems are designed for controlled environments and require further adaptation for practical industrial deployment.

He et al. conducted a comprehensive review on computer vision techniques used for food quality evaluation and reported that image-based analysis can effectively identify defects,

discoloration, and structural variations in food products[8]. However, implementing such systems in real industrial environments requires further integration with automated production systems.

Kamilaris and Prenafeta-Boldú presented a survey on deep learning applications in agriculture and food industries, highlighting how convolutional neural networks can significantly improve image-based classification tasks[9]. Their findings suggest that deep learning techniques can enhance automation and decision-making processes in food quality monitoring systems.

In addition, Tan and Le introduced EfficientNet, a deep learning architecture designed to improve image classification accuracy while reducing computational complexity[10]. Such architectures have shown strong potential in computer vision applications, including food defect detection and quality grading.

Finally, Redmon and Farhadi developed the YOLO object detection framework, which enables real-time object detection with high accuracy and efficiency[11]. YOLO-based models have been widely adopted in industrial inspection systems due to their ability to detect defects and classify objects in real time.

Although these studies demonstrate significant progress in AI-based food inspection technologies, several limitations remain. Most existing research focuses on isolated aspects of quality assessment such as defect detection or freshness classification. Few systems integrate computer vision with automated sorting mechanisms to provide a comprehensive assessment of raw fish quality before processing. In addition, many proposed solutions are developed under controlled laboratory conditions and are not designed for real-time deployment in small-scale seafood processing environments.

Therefore, there is a need for an integrated system that combines computer vision techniques with automated sorting mechanisms to provide accurate and standardized raw fish quality assessment in real industrial processing environments.

1.3 Research Gap

Although recent research has demonstrated the effectiveness of artificial intelligence and computer vision techniques in food quality inspection, several critical limitations remain in existing seafood quality assessment systems. Most current approaches focus on isolated aspects of fish quality analysis rather than providing a comprehensive solution for evaluating raw fish before processing. For example, many computer vision studies primarily analyze visual features such as texture, color, or surface patterns to predict fish freshness or detect defects [3], [5]. While these methods can improve inspection accuracy compared to manual grading, they are often limited to laboratory-based experimental setups and do not fully address the requirements of real industrial processing environments.

Another limitation of existing research is that many proposed systems are developed using controlled laboratory datasets rather than real industrial processing conditions. In practice, fish processing environments involve variable lighting conditions, irregular fish sizes, and high-

speed conveyor operations. Systems designed in laboratory environments may therefore struggle to maintain consistent performance when deployed in real production facilities. Furthermore, several studies focus on defect detection or freshness classification but do not integrate automated decision-making mechanisms capable of supporting real-time industrial sorting processes [2], [6].

In addition, most available technological solutions emphasize environmental monitoring or post-processing quality analysis rather than addressing the critical stage of raw fish intake evaluation. Previous studies have explored computer vision techniques for detecting seafood defects and evaluating fish freshness [4], [7], but these approaches typically analyze isolated quality indicators without integrating them into an automated grading and sorting workflow suitable for industrial processing lines.

Another significant gap is the lack of affordable and scalable automation solutions specifically designed for small- and medium-scale seafood processing facilities. Many advanced food inspection technologies are developed for large industrial manufacturing systems and require expensive equipment and complex infrastructure. These solutions are often impractical for traditional Maldive fish processing operations that operate with limited technological resources. Research also indicates that many machine vision systems in food inspection are designed for controlled environments and require further adaptation for real-world industrial deployment [8], [9].

Therefore, there is a clear need for an integrated system that combines **computer vision techniques with automated sorting mechanisms** to perform accurate and standardized raw fish quality assessment in real-time processing environments. The proposed research addresses this gap by developing an **AI-driven raw fish quality assessment and standardization system** that uses RGB camera-based computer vision to automatically detect defects, evaluate visual freshness indicators, and classify fish into standardized grading categories such as **Grade A, Grade B, or Reject**. By integrating automated quality classification with conveyor-based sorting mechanisms, the proposed system aims to improve processing efficiency, reduce human error, and introduce scalable intelligent automation into traditional seafood processing industries.

Comparison of the Existing Research Methods and the Proposed Method

Technologies and Methods	**[2]**	**[3]**	**[4]**	**[5]**	**[6]**	**[7]**	**[8]**	**[9]**	**Proposed Method**
Computer vision for fish freshness detection	✓	✓	✗	✓	✓	✓	✓	✓	✓
Detection of fish surface defects	✓	✗	✗	✓	✓	✓	✓	✓	✓
Use of RGB camera image analysis	✓	✓	✗	✓	✓	✓	✓	✓	✓
Automated fish quality grading	✓	✓	✗	✓	✗	✗	✗	✗	✓
Real-time processing capability	✗	✗	✗	✗	✗	✗	✗	✗	✓
Integration with an automated sorting system	✗	✗	✗	✗	✗	✗	✗	✗	✓
Suitability for small-scale seafood industries	✗	✗	✓	✗	✗	✗	✗	✗	✓
Comprehensive fish quality evaluation (defect + freshness)	✗	✓	✗	✓	✗	✗	✗	✗	✓

2. OBJECTIVES

2.1 Main Objective

To design and develop an **AI-driven raw fish quality assessment and standardization system using computer vision techniques** that can automatically detect surface defects, evaluate visual freshness indicators, and classify raw fish into standardized quality grades with high accuracy before processing.

2.2 Specific Objectives

1. **To collect and prepare a dataset of raw fish images** representing different quality conditions, such as fresh, defective, and substandard fish samples.
2. **To preprocess and annotate the collected image dataset** to support supervised deep learning model training.
3. **To develop a computer vision model using deep learning techniques** to detect visual defects and evaluate quality indicators from RGB images of raw fish.
4. **To design and implement an automated fish classification system** that categorizes fish into predefined quality grades such as Grade A, Grade B, or Reject.
5. **To integrate the classification system with a conveyor-based sorting mechanism** for real-time quality-based fish sorting in processing environments.
6. **To evaluate the performance of the proposed system** using metrics such as classification accuracy, precision, recall, and processing time.

3. METHODOLOGY

This study adopts a **Design Science Research** (DSR) approach, which focuses on designing and developing a technological artifact to solve a real-world problem. The objective of this research is to develop an AI-driven raw fish quality assessment and standardization system that improves the accuracy and consistency of raw fish inspection in Maldive fish processing facilities.

The design science approach is appropriate because the research involves creating and evaluating a functional system consisting of computer vision models, automated classification mechanisms, and real-time sorting support. The research process includes problem identification, system design, dataset preparation, model development, prototype implementation, and performance evaluation.

Technical Methods and Algorithms

The proposed system utilizes computer vision techniques and deep learning algorithms to evaluate raw fish quality before processing.

First, image acquisition will be performed using a high-resolution RGB camera installed above the intake conveyor belt. The captured images will represent different fish quality conditions including fresh fish, defective fish, and substandard samples.

Next, image preprocessing techniques will be applied to improve data quality before model training. These techniques include image resizing, normalization, background removal, and noise reduction using image processing libraries. These steps help standardize the dataset and improve model performance.

Following preprocessing, deep learning algorithms will be used to detect defects and classify fish quality. Convolutional Neural Networks (CNNs) will be applied for feature extraction and image classification tasks. Additionally, object detection architectures such as YOLO may be used to identify visible surface defects such as bruises, discoloration, and physical damage on fish surfaces.

After feature extraction and defect detection, a classification model will categorize fish into predefined quality grades such as Grade A, Grade B, or Reject. The classification output will be connected to a conveyor-based sorting mechanism that automatically directs fish into appropriate processing channels according to their quality grade.

Tools and Platforms

Several software tools and development platforms will be used to implement the proposed system.

- **Python Programming Language** – Used as the primary development environment due to its extensive support for machine learning and image processing frameworks.
- **OpenCV** – Used for image preprocessing tasks such as resizing, filtering, and image transformation.
- **TensorFlow / PyTorch** – Used for developing and training deep learning models for fish quality classification.
- **Ultralytics YOLO Framework** – Used for real-time defect detection and object detection tasks.
- **Google Colab** – Used for model training, experimentation, and performance evaluation.

These tools are selected because they provide robust frameworks for developing AI-based computer vision systems and allow efficient experimentation with deep learning models.

Data Collection and Ethical Considerations

Data collection will involve capturing images of raw fish samples under different quality conditions using an RGB camera. The dataset will include various fish samples representing different levels of freshness, defects, and physical conditions.

Images will be manually annotated and labeled according to predefined quality categories such as Grade A, Grade B, and Reject. These labeled datasets will be used for supervised machine learning model training.

Ethical considerations will be addressed by ensuring that data collection is conducted with proper authorization from processing facilities. The collected data will be used strictly for academic research purposes and will not contain any confidential or sensitive industrial information.

Validation Strategy and Performance Metrics

To evaluate the effectiveness of the proposed system, the dataset will be divided into training, validation, and testing datasets. Machine learning models will be trained using the training dataset and evaluated using the testing dataset.

The performance of the system will be measured using several evaluation metrics commonly used in machine learning and classification tasks.

Key performance metrics include:

- **Accuracy**: Percentage of correctly classified fish samples.
- **Precision**: Proportion of correct positive predictions for each quality class.
- **Recall**: Ability of the model to correctly identify defective or low-quality fish samples.
- **F1-score**: Harmonic mean of precision and recall to evaluate classification balance.
- **Confusion Matrix**: Visualization of classification performance across quality categories.

In addition to classification performance, system processing time and detection speed will also be evaluated to determine the feasibility of deploying the system in real-time industrial processing environments.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system introduces an AI-driven automated raw fish quality assessment and standardization framework designed to improve the accuracy and efficiency of quality inspection in Maldive fish processing facilities. The system integrates computer vision technology, deep learning models, and automated decision-making mechanisms to evaluate raw fish before they enter the processing stage.

The system operates by capturing images of raw fish using an RGB camera installed above the intake conveyor belt. These images are analyzed using a trained deep learning model to identify visible defects and assess visual indicators of fish quality such as discoloration, bruising, and surface damage.

The AI model processes the captured images and classifies fish into predefined quality categories such as Grade A, Grade B, or Reject. Based on the classification result, the system triggers an automated sorting mechanism on the conveyor belt, directing fish to the appropriate processing channel. A monitoring dashboard displays classification results, system alerts, and quality statistics in real time.

This architecture enables consistent quality assessment, reduced human error, and improved operational efficiency in seafood processing environments.

4.2 System Components

The high-level system architecture consists of the following major components:

1. Image Acquisition Module

- RGB camera captures images of raw fish moving on the intake conveyor belt.

2. Data Processing Module

- Performs image preprocessing tasks such as resizing, normalization, and noise reduction.

3. AI Classification Module (Individual Component Focus)

- Deep learning model analyzes images to detect visual defects and evaluate fish quality.
- Classifies fish into predefined quality categories.

4. Decision and Sorting Module

- Sends control signals to the conveyor sorting mechanism based on classification results.

5. Monitoring and Dashboard Module

- Displays classification results, system statistics, and operational alerts.

4.3 Individual Component Highlight

The primary focus of this research is the AI-Driven Raw Fish Quality Assessment Module. This component performs the core intelligence of the system by analyzing visual features extracted from fish images and generating quality classifications.

Key subcomponents of this module include:

- Image preprocessing subsystem
- Feature extraction using deep learning models
- Defect detection algorithm
- Quality classification model
- Quality scoring and grading mechanism

This module transforms raw image inputs into actionable quality decisions that drive the automated sorting process.

4.4 Component Interactions

The components of the system interact in a sequential processing pipeline.

1. Raw fish enters the conveyor belt.
2. The RGB camera captures an image of the fish.
3. The captured image is sent to the data processing module.
4. The AI classification module analyzes the image and predicts the fish quality category.
5. The decision module sends control signals to the conveyor sorting system.
6. The fish is automatically directed to the correct sorting channel.
7. System results are recorded and displayed on the monitoring dashboard.

This coordinated interaction ensures smooth integration between image capture, AI analysis, and automated processing.

4.5 Data Flow Explanation

The system processes data through several stages:

1. Data Acquisition

- Fish images are captured using an RGB camera.

2. Data Preprocessing

- Images are cleaned, resized, and normalized before being sent to the AI model.

3. Feature Extraction

- Deep learning models extract visual features from fish images.

4. Quality Classification

- The AI model predicts fish quality categories such as Grade A, Grade B, or Reject.

5. Decision Execution

- Classification results trigger automated sorting actions on the conveyor system.

6. Data Logging and Visualization

- Results are stored and displayed through the monitoring dashboard.

This pipeline enables real-time AI-based inspection and automated decision-making in fish processing operations.

4.6 Scalability and Security Considerations

The system is designed with scalability in mind to allow deployment across different seafood processing facilities. Additional cameras or processing units can be integrated to handle larger production volumes without significant system redesign.

From a security perspective, collected data will be stored in secure databases to prevent unauthorized access. Communication between system components will use secure protocols to maintain data integrity and system reliability.

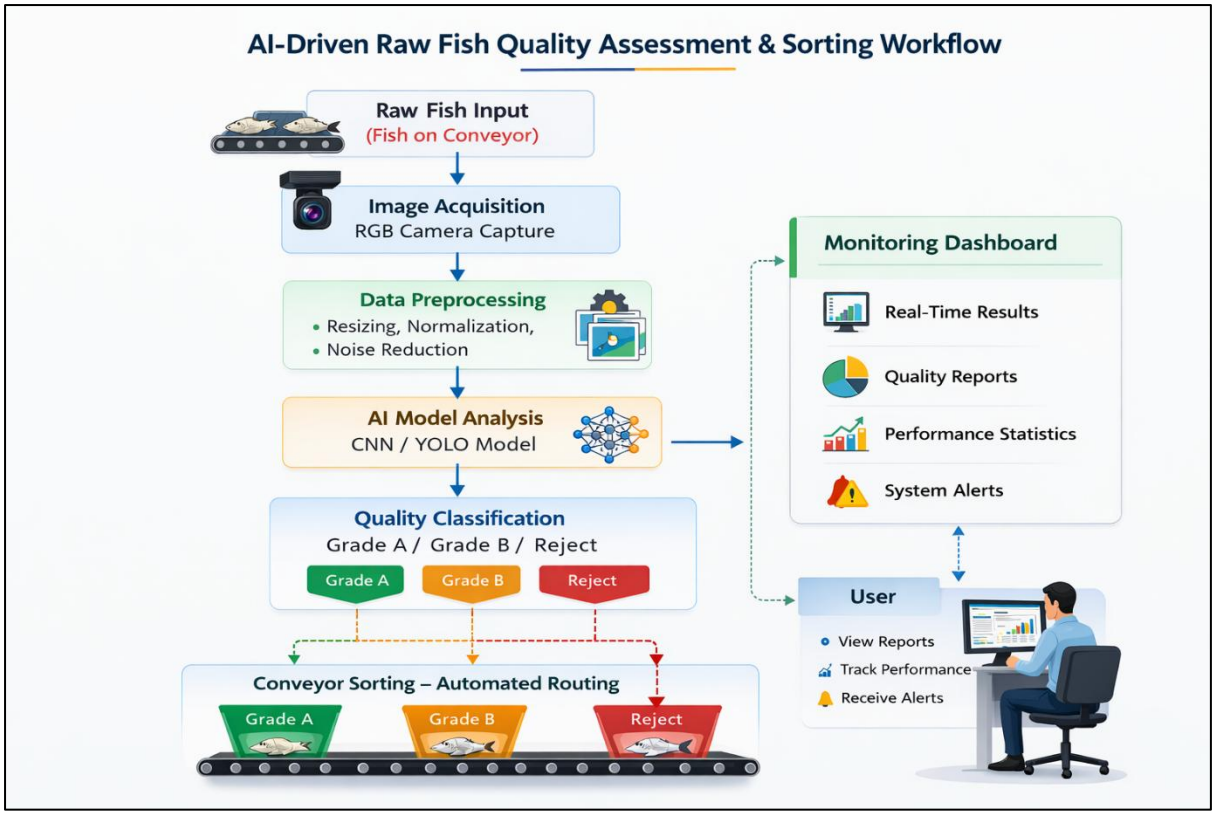


Figure 1 - High-level system architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed AI-driven raw fish quality assessment and sorting system interacts with several stakeholders involved in the Maldive fish processing workflow. Each stakeholder has specific needs and expectations from the system.

Table 1 - List of Stakeholders

Stakeholder	Role	Needs
Fish Processing Operators	Handle raw fish intake and sorting	Need fast and accurate quality inspection to reduce manual workload
Quality Assurance Officers	Monitor product quality standards	Require reliable grading results and defect detection
Processing Plant Managers	Oversee production efficiency	Need real-time monitoring and analytics for decision making
Exporters and Distributors	Ensure product quality for international markets	Require consistent grading and reduced defect rates
System Administrators / Technicians	Maintain and manage the system	Need reliable operation and easy system management
System Users / Operators	Interact with the dashboard and sorting system	Need clear system alerts, reports, and simple system controls

5.2 Functional Requirements

The functional requirements describe the operations the proposed system must perform.

FR1: The system shall capture high-resolution images of raw fish using an RGB camera installed above the intake conveyor.

FR2: The system shall preprocess captured images using techniques such as resizing, normalization, and noise reduction.

FR3: The system shall analyze fish images using a deep learning model to detect surface defects such as bruises, discoloration, or deformities.

FR4: The system shall classify fish into predefined quality categories such as Grade A, Grade B, or Reject.

FR5: The system shall send classification results to the conveyor sorting mechanism for automated fish routing.

FR6: The system shall store captured images and classification results for monitoring and future analysis.

FR7: The system shall display quality results and system status on a monitoring dashboard.

FR8: The system shall generate alerts when defective or substandard fish are detected.

FR9: The system shall allow system users to monitor system performance and view inspection reports through the dashboard interface.

5.3 Non-Functional Requirements

Non-functional requirements define system performance, reliability, and operational constraints.

Performance Requirements

- The system should process and classify each fish within 2–3 seconds to support real-time conveyor operations.
- The AI model should achieve at least 90% classification accuracy for fish quality grading.

Security Requirements

- Data collected by the system should be stored securely to prevent unauthorized access.
- Access to the monitoring dashboard should be restricted to authorized users only.

Usability Requirements

- The system dashboard should provide clear visual indicators and alerts for operators.
- The user interface should be simple and easy to use for non-technical staff.

Reliability Requirements

- The system should operate continuously during processing hours with minimal downtime.
- The camera system should provide stable image acquisition under industrial lighting conditions.

Scalability Requirements

- The system should support integration with additional cameras or processing units in the future.
- The architecture should allow deployment in multiple processing facilities.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system includes small- and medium-scale Maldive fish processing facilities in Sri Lanka and the Maldives. These processors commonly rely on manual inspection and traditional processing methods, which often result in inconsistent quality grading and operational inefficiencies. The proposed system provides an affordable and scalable automation solution to improve quality control and production efficiency.

In addition to fish processors, other potential customers include seafood exporters, food processing companies, and food quality assurance laboratories that require accurate fish quality inspection tools.

A typical customer persona is a processing plant manager or owner of a seafood processing facility who aims to reduce labor dependency, improve product consistency, and comply with international seafood export standards.

6.2 Value Proposition

The proposed system offers several key benefits to seafood processing industries:

- Automated fish quality inspection that replaces subjective manual grading.
- Higher accuracy and consistency in detecting fish defects.
- Improved processing efficiency by reducing inspection time.
- Reduced operational costs by minimizing labor dependency.
- Better product quality consistency, helping processors meet export quality standards.
- A real-time monitoring dashboard that supports operational and quality management decisions.

By introducing AI-driven automation into the raw fish intake stage, the system helps processors reduce waste, improve throughput, and maintain standardized product quality.

6.3 Cost Estimation

The estimated cost of developing and deploying the prototype system includes hardware components and system integration.

Table 2 - Cost Estimation

Component	Estimated Cost (LKR)
High-Resolution RGB Camera	LKR 10,000 – 12,000
Edge Processing Unit (Raspberry Pi)	LKR 25,000 – 35,000
Lighting System for Image Capture	LKR 6,000 – 8,000
Software Development	LKR 0 (Open-source tools)

Using open-source tools such as Python, TensorFlow, and OpenCV significantly reduces the development cost.

6.4 Competitive Advantage

The proposed system provides several competitive advantages compared to existing solutions:

1. Industry-specific design tailored for Maldive fish processing operations.
2. AI-driven computer vision inspection, which improves grading accuracy.
3. Automated conveyor sorting capability, rarely used in traditional seafood processing.
4. Affordable hardware design, making it accessible to small and medium processors.
5. Real-time monitoring and analytics dashboard for operational insights.

These features allow the system to deliver cost-effective intelligent automation compared with expensive industrial inspection systems.

6.5 Intellectual Property Considerations

The intellectual property of the proposed system may involve several components:

- AI models and classification algorithms developed for fish quality inspection may be protected through software copyright.
- System architecture and automation workflow could be considered for patent protection if the design introduces a novel industrial automation mechanism.
- Training datasets and trained AI models may be treated as proprietary research assets.

If the system demonstrates strong commercial potential, it could be registered as a software product or patented industrial technology, enabling commercialization and potential startup development.

7. BUDGET AND JUSTIFICATION

The proposed system requires several hardware components, software tools, and development effort to implement the **AI-driven raw fish quality assessment and sorting system**. Most of the software tools used in this project are **open-source**, which significantly reduces development costs.

7.1 Hardware Costs

The hardware components are required to **capture fish images, process AI models, and control the conveyor sorting mechanism**.

Table 3 - Hardware Costs

Hardware Component	Quantity	Estimated Cost (LKR)	Purpose
High-Resolution RGB Camera	1	LKR 12,000	Capture fish images for AI analysis
Edge Processing Unit (Raspberry Pi)	1	LKR 30,000	Run AI inference and data processing
LED Lighting Setup	1	LKR 3,000	Ensure stable image capture conditions
Conveyor Frame / Mounting Components	1	LKR 8,000	Support camera and conveyor installation

Total Hardware Cost: LKR 53,000

7.2 Software Licenses

Most software tools used in this research are **open-source platforms**, eliminating licensing costs.

Table 4 - Software Licenses

Software Tool	License Cost	Purpose
Python	Free	Main development language
OpenCV	Free	Image processing
PyTorch	Free	Deep learning model development
Ultralytics YOLO Framework	Free	Object detection
Google Colab	Free	Model training and experimentation

Total Software Cost: LKR 0

7.3 Justification of Budget

The selected hardware components are essential for implementing the AI-driven raw fish quality assessment system. The **high-resolution RGB camera** is required to capture clear images of raw fish so that the computer vision model can accurately detect visual defects and quality indicators. The **edge processing unit (Raspberry Pi)** provides sufficient computational capability to process captured images and run the trained AI model in real time. The **LED lighting setup** ensures consistent illumination during image acquisition, which improves image quality and enhances the accuracy of the AI model. The **conveyor frame and mounting components** are necessary to securely position the camera and support the inspection environment within the processing workflow.

Most of the software tools used in the system are **open-source platforms**, which eliminates licensing costs and makes the solution more cost-effective. Programming languages and libraries such as **Python, OpenCV, and PyTorch/TensorFlow** provide powerful capabilities for image processing and deep learning model development. Additionally, tools such as the **Ultralytics YOLO framework** enable efficient object detection, while **Google Colab** provides cloud-based computational resources for model training and experimentation. Using open-source technologies allows the system to be developed and deployed at a lower cost while maintaining high technical capability.

8. WORK BREAKDOWN STRUCTURE (WBS)

This work breakdown structure outlines the major tasks required to design, develop, and evaluate the **AI-driven raw fish quality assessment system**. The tasks are organized into several phases, including **research, dataset preparation, model development, system integration, testing, and documentation**.

8.1 Task Breakdown

Table 5 - Work breakdown structure

WBS ID	Task	Description
1.0	Literature Review	Review research papers related to fish freshness detection, computer vision, and AI-based food inspection
1.1	Requirement Analysis	Identify system requirements for raw fish quality assessment
2.0	Dataset Collection	Capture raw fish images representing different quality conditions
2.1	Data Annotation	Label collected images according to quality categories (Grade A, Grade B, Reject)
3.0	Image Preprocessing	Perform image cleaning, resizing, normalization, and noise reduction
3.1	Feature Extraction	Extract visual features using deep learning techniques
4.0	AI Model Development	Develop a deep learning model for fish quality classification
4.1	Model Training	Train the model using the prepared dataset
4.2	Model Evaluation	Evaluate model performance using accuracy, precision, recall, and F1-score
5.0	System Integration	Integrate RGB camera input with the AI model and processing system
6.0	Automated Sorting Logic	Implement classification output to control the conveyor sorting mechanism
7.0	System Testing	Test the full system with real fish samples
7.1	Performance Evaluation	Measure system accuracy, processing speed, and reliability
8.0	Documentation	Prepare project documentation and technical reports

8.2 Workload Distribution

The workload is distributed across different development phases to ensure steady progress throughout the project timeline.

Research and Planning Phase:
Focuses on literature review, problem analysis, and system requirement identification.

Development Phase:
Includes dataset preparation, image preprocessing, AI model development, and training.

Integration Phase:
Involves connecting the AI model with the camera system and conveyor sorting logic.

Evaluation Phase:
Focuses on testing system performance and validating classification accuracy.

Final Phase:
Includes documentation, report writing, and preparation for project evaluation.

This structured approach ensures that all tasks required for the **AI-driven fish quality assessment module** are completed systematically within the project timeline.

9. GANTT CHAR

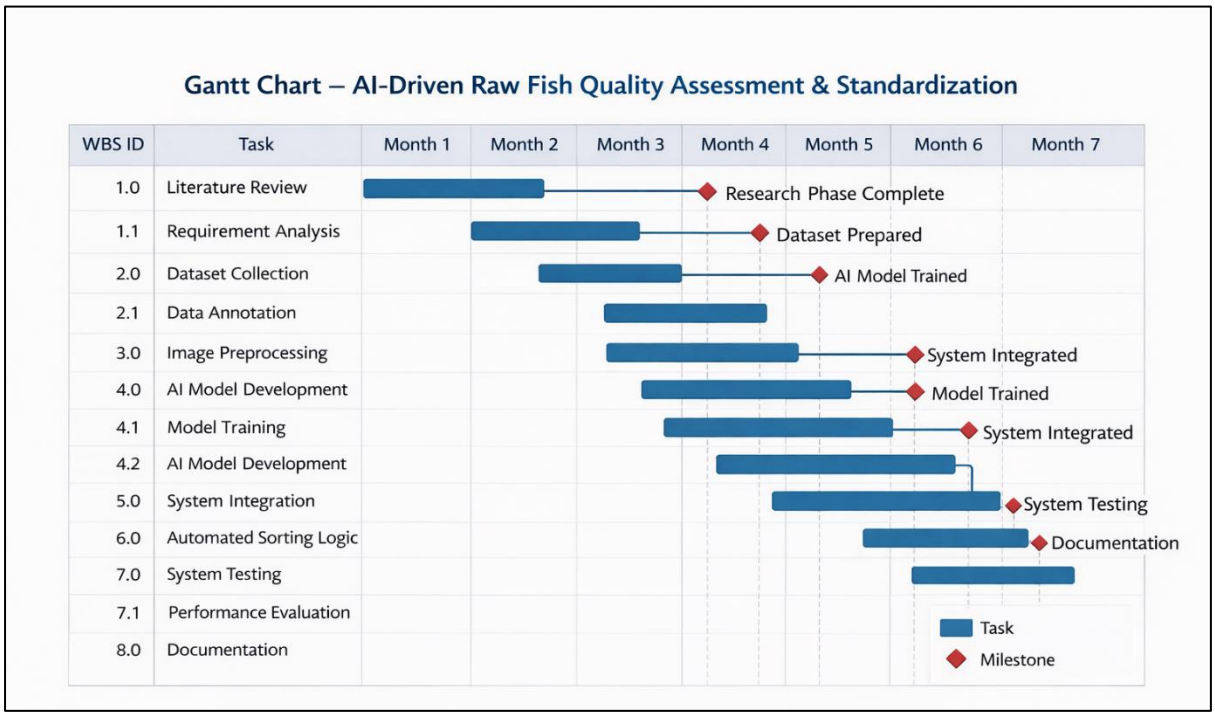


Figure 2 – Gantt Chart

10. REFERENCES LIST

[1] R. A. M. Nafchi et al., “Applications of IoT and machine learning in smart food processing,” Food Engineering Reviews, 2021.

[2] S. S. Mir et al., “AI-enabled quality assessment in fish processing,” IEEE Access, 2022.

[3] Y. Cheng et al., “Computer vision for fish freshness evaluation,” IEEE Access, 2022.

[4] P. Kulatunga et al., “Challenges in traditional dried fish industry in South Asia,” Journal of Food Processing, 2020.

[5] S. Liu et al., “Computer vision-based quality assessment of fish products using deep convolutional neural networks,” Computers and Electronics in Agriculture, 2022.

[6] A. Das et al., “Deep learning for seafood defect detection,” Computers and Electronics in Agriculture, 2021.

[7] J. Wang et al., “Machine vision systems for food quality inspection: A review,” Food Engineering Reviews, 2020.

[8] Y. He et al., “Computer vision for food quality evaluation: A review,” Trends in Food Science & Technology, 2017.

[9] A. Kamilaris and F. X. Prenafeta-Boldú, “Deep learning in agriculture: A survey,” Computers and Electronics in Agriculture, 2018.

[10] M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in Proc. ICML, 2019.

[11] J. Redmon and A. Farhadi, “YOLOv3: An incremental improvement,” arXiv preprint arXiv:1804.02767, 2018.

11. APPENDICES

The appendices provide additional supporting materials related to the proposed research system. These include survey instruments, interface mockups, risk analysis, and ethical considerations that support the development and implementation of the **AI-driven raw fish quality assessment system**.

Appendix A – Survey / Interview Instruments

To understand current practices in Maldive fish processing, brief interviews or surveys may be conducted with fish processing operators and quality control personnel.

Example Questions

1. How is raw fish quality currently inspected in your processing facility?
2. What challenges do you face with manual fish grading and inspection?
3. How much time does manual fish inspection take during intake processing?
4. What types of defects are most commonly observed in raw fish?
5. Would automated quality inspection systems improve your processing efficiency?

The responses will help identify real-world operational challenges and validate the need for an **AI-based automated inspection system**.

Appendix B – Preliminary User Interface Mockups

The monitoring dashboard will allow operators to observe system performance and fish quality results in real time.

Key UI Features

- Real-time camera feed of fish inspection
- Fish quality classification results (**Grade A / Grade B / Reject**)
- Defect detection alerts for damaged or low-quality fish
- System performance statistics and inspection logs
- Quality analytics reports for batch monitoring

The dashboard will be designed to provide a **simple and intuitive interface** suitable for non-technical users in fish processing environments.

Appendix C – Risk Analysis

The project may encounter several technical and operational risks during development and implementation.

Risk	Description	Mitigation Strategy
Insufficient training data	Limited fish image dataset may affect model accuracy	Collect images from multiple sources and apply data augmentation
Lighting variations	Poor lighting conditions may reduce image quality	Use controlled LED lighting near the camera
AI model overfitting	Model may perform well on training data but poorly on new data	Use validation datasets and cross-validation
Hardware integration issues	Difficulty integrating camera and conveyor control system	Conduct incremental testing and debugging
Real-time processing delay	AI inference may slow conveyor operations	Optimize model and use edge computing hardware

Appendix D – Ethical Considerations

The proposed research does not involve human subjects, personal data, or medical experimentation. The study focuses on industrial processing of seafood products and the development of an **AI-based quality inspection system**.

However, the following ethical guidelines will be maintained:

- Data collected will be used only for **academic research purposes**.
- Any collaboration with processing facilities will be conducted with **organizational consent**.
- The system will not collect **personal or sensitive information**.
- Data collected from fish processing environments will be stored securely.

Therefore, the project is expected to have **minimal ethical concerns**.